

Homework 01

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Problem 1. *Let E and F be events. Using the axioms of probability, prove that*

$$P(E \cup F) = P(E) + P(F) - P(EF).$$

E is carved by F into two disjoint pieces:

$$E = EF \cup EF^c \quad (\text{disjoint}),$$

so by axiom 3,

$$P(E) = P(EF) + P(EF^c) \iff P(EF^c) = P(E) - P(EF).$$

Similarly, $P(FE^c) = P(F) - P(EF)$.

Therefore, since $E \cup F = EF \cup EF^c \cup FE^c$ is a union of pairwise disjoint sets,

$$\begin{aligned} P(E \cup F) &= P(EF \cup EF^c \cup FE^c) \\ &= P(EF) + P(EF^c) + P(FE^c) \\ &= P(EF) + (P(E) - P(EF)) + (P(F) - P(EF)) \\ &= P(E) + P(F) - P(EF). \end{aligned}$$

Problem 8. *If $P(E) = 0.9$ and $P(F) = 0.8$, show that $P(EF) \geq 0.7$. In general, show that*

$$P(EF) \geq P(E) + P(F) - 1.$$

This is known as Bonferroni's inequality.

First we record the bound the argument rests on.

Lemma. *For any event A , $P(A) \leq 1$.*

Since $\Omega = A \cup A^c$ is a union of disjoint sets,

$$P(\Omega) = 1 = P(A \cup A^c) = P(A) + P(A^c) \quad (\text{disjoint}),$$

so

$$P(A) + P(A^c) = 1 \iff P(A) = 1 - P(A^c) \implies P(A) \leq 1,$$

since $P(A^c) \geq 0$ by the first axiom.

Now prove $P(EF) \geq P(E) + P(F) - 1$. Recall, using Problem 1,

$$1 \geq P(E \cup F) = P(E) + P(F) - P(EF),$$

and therefore

$$1 + P(EF) \geq P(E) + P(F) \iff P(EF) \geq P(E) + P(F) - 1.$$

Therefore, if $P(E) = 0.9$ and $P(F) = 0.8$,

$$P(EF) \geq 0.9 + 0.8 - 1 \geq 0.7.$$

Problem 10. Show that

$$P\left(\bigcup_{i=1}^n E_i\right) \leq \sum_{i=1}^n P(E_i).$$

This is known as Boole's inequality.

Hint: Either use Equation (1.2) and mathematical induction, or else show that $\bigcup_{i=1}^n E_i = \bigcup_{i=1}^n F_i$, where $F_1 = E_1$, $F_i = E_i \cap \bigcap_{j=1}^{i-1} E_j^c$, and use property (iii) of a probability.

First we record the monotonicity of P , which the argument rests on.

Lemma. If $A \subseteq B$ then $P(A) \leq P(B)$.

Since $A \subseteq B$, we may carve $B = A \cup BA^c$, a union of disjoint sets, so

$$P(B) = P(A \cup BA^c) = P(A) + P(BA^c) \quad (\text{disjoint}).$$

Note $P(BA^c) \geq 0$ by the first axiom, so

$$P(A) = P(B) - P(BA^c) \leq P(B).$$

Now prove Boole's inequality. Let

$$F_1 = E_1, \quad F_i = E_i E_1^c E_2^c \cdots E_{i-1}^c \quad (i \geq 2).$$

Note each $F_i \subseteq E_i$, and F_i is defined such that $\bigcup_{i=1}^n E_i = \bigcup_{i=1}^n F_i$. Further, $i \neq j \implies F_i \cap F_j = \emptyset$. So

$$P\left(\bigcup_{i=1}^n E_i\right) = P\left(\bigcup_{i=1}^n F_i\right) = \sum_{i=1}^n P(F_i) \quad (\text{disjoint}).$$

Since $F_i \subseteq E_i$ for all i , the Lemma gives $P(F_i) \leq P(E_i)$, and therefore $\sum_{i=1}^n P(F_i) \leq \sum_{i=1}^n P(E_i)$. Therefore

$$P\left(\bigcup_{i=1}^n E_i\right) \leq \sum_{i=1}^n P(E_i).$$

Aside. Clearly $\bigcup_{i=1}^n F_i \subseteq \bigcup_{i=1}^n E_i$, since $F_i \subseteq E_i$ for all i . It remains to prove the other inclusion, $\bigcup_{i=1}^n E_i \subseteq \bigcup_{i=1}^n F_i$, which we do by induction on n .

Base case. For $n = 1$, $\bigcup_{i=1}^1 E_i = E_1 = F_1 = \bigcup_{i=1}^1 F_i$.

Inductive step. Assume $\bigcup_{i=1}^{n-1} E_i \subseteq \bigcup_{i=1}^{n-1} F_i$. Let $e \in \bigcup_{i=1}^n E_i$. If $e \in F_n$ we are done. If $e \notin F_n = E_n E_1^c \cdots E_{n-1}^c$, then either $e \notin E_n$, or there exists $i < n$ such that $e \in E_i$.

If the latter is true, then

$$e \in \bigcup_{i=1}^{n-1} E_i \subseteq \bigcup_{i=1}^{n-1} F_i \subseteq \bigcup_{i=1}^n F_i,$$

using the inductive hypothesis. If the former is true, the latter must still be true, since $e \in \bigcup_{i=1}^n E_i$ and $e \notin E_n$ force $e \in E_i$ for some $i < n$.

Problem 12. Let E and F be mutually exclusive events in the sample space of an experiment. Suppose that the experiment is repeated until either event E or event F occurs. What does the sample space of this new super experiment look like? Show that the probability that event E occurs before event F is $P(E)/[P(E) + P(F)]$.

Hint: Argue that the probability that the original experiment is performed n times and E appears on the n th time is $P(E) \times (1 - p)^{n-1}$, $n = 1, 2, \dots$, where $p = P(E) + P(F)$. Add these probabilities to get the desired answer.

An outcome of the super experiment is a finite sequence $(o_1, o_2, \dots, o_\ell)$ of outcomes of the original experiment, where $\ell \in \mathbb{N}$, $o_i \in E^c F^c$ for every $i < \ell$, and $o_\ell \in E \cup F$. The sample space of the super experiment is the set of all such sequences.

For each $n \geq 1$, the original experiment is performed n times and E appears on the n th time exactly when the first $n - 1$ trials produce neither E nor F and the n th produces E , so

$$P(E \text{ before } F \text{ on trial } n) = P(E^c F^c)^{n-1} P(E).$$

These events are mutually exclusive across n , so by the countable additivity axiom,

$$\begin{aligned} P(E \text{ before } F) &= P\left(\bigcup_{n=1}^{\infty} \{E \text{ before } F \text{ on trial } n\}\right) && \text{(axiom } \rightarrow \text{ mutually exclusive)} \\ &= \sum_{n=1}^{\infty} P(E^c F^c)^{n-1} P(E) && \text{(geometric)} \\ &= \frac{P(E)}{1 - P(E^c F^c)} = \frac{P(E)}{P(E \cup F)} = \frac{P(E)}{P(E) + P(F)} && \text{(mutually exclusive } \rightarrow \text{ axiom).} \end{aligned}$$

Problem 39. Stores A , B , and C have 50, 75, and 100 employees and, respectively, 50, 60, and 70 percent of these are women. Resignations are equally likely among all employees, regardless of sex. One employee resigns and this is a woman. What is the probability that she works in store C ?

Write C for the event that the employee who resigns works in store C , and W for the event that the employee who resigns is a woman. By the definition of conditional probability and then the multiplication rule,

$$P(C | W) = \frac{P(CW)}{P(W)} = \frac{P(C) P(W | C)}{P(W)}.$$

The resignation is equally likely among all $50 + 75 + 100 = 225$ employees, so $P(C) = 100/225$, and $P(W | C) = 0.7$. Splitting W across the three stores gives the denominator, so

$$P(C | W) = \frac{\frac{100}{225} \cdot 0.7}{0.5 \cdot \frac{50}{225} + 0.6 \cdot \frac{75}{225} + 0.7 \cdot \frac{100}{225}} = \frac{0.7 \cdot \frac{4}{9}}{0.5 \cdot \frac{2}{9} + 0.6 \cdot \frac{1}{3} + 0.7 \cdot \frac{4}{9}} = \frac{\frac{14}{45}}{\frac{1}{9} + \frac{1}{5} + \frac{14}{45}}.$$

$$\text{Since } \frac{1}{9} + \frac{1}{5} = \frac{14}{45},$$

$$P(C | W) = \frac{14/45}{14/45 + 14/45} = \frac{1}{2}.$$

Problem 41. In a certain species of rats, black dominates over brown. Suppose that a black rat with two black parents has a brown sibling.

- (a) What is the probability that this rat is a pure black rat (as opposed to being a hybrid with one black and one brown gene)?
- (b) Suppose that when the black rat is mated with a brown rat, all five of their offspring are black. Now, what is the probability that the rat is a pure black rat?

(a) Write B for the black gene and b for the brown one, so that a rat is BB , Bb or bb and is black exactly when it carries at least one B .

The brown sibling is bb , so each parent handed down a b ; both parents are black, so each also carries a B . Both parents are therefore Bb . The four gene pairs BB , Bb , bB , bb an offspring of $Bb \times Bb$ may receive are equally likely, and conditioning on the rat itself being black leaves the three pairs BB , Bb , bB . Hence

$$P(BB \mid bb \text{ sibling, parents } Bb \text{ and } Bb, \text{ self black}) = \frac{1}{3}.$$

(b) By the definition of conditional probability and the multiplication rule,

$$P(BB \mid 5 \text{ black offspring with a } bb \text{ mate}) = \frac{P(BB) P(5 \text{ black offspring with a } bb \text{ mate} \mid BB)}{P(5 \text{ black offspring with a } bb \text{ mate})}.$$

For the denominator, split over the two possibilities for the rat, whose probabilities $1/3$ and $2/3$ come from part (a). A BB rat mated with a bb rat produces only Bb offspring, all black; a Bb rat mated with a bb rat produces a black offspring with probability $1/2$ each time. Hence

$$\begin{aligned} P(5 \text{ black offspring}) &= P(5 \text{ black offspring} \mid BB) P(BB) + P(5 \text{ black offspring} \mid Bb) P(Bb) \\ &= (1)^5 \left(\frac{1}{3}\right) + \left(\frac{1}{2}\right)^5 \left(\frac{2}{3}\right) = \frac{1}{3} + \frac{2}{96} = \frac{16}{48} + \frac{1}{48} = \frac{17}{48}. \end{aligned}$$

Therefore

$$P(BB \mid 5 \text{ black offspring with a } bb \text{ mate}) = \frac{\frac{1}{3} \cdot (1)^5}{\frac{17}{48}} = \frac{48}{51} = \frac{16}{17}.$$